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Research Article

Extracts of *Cordyceps militaris* Lower Blood Glucose via the Stimulation of Cholinergic Activation and Insulin Secretion in Normal Rats

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Abstract

Previous studies have shown that *Cordyceps militaris* (CM) has a hypoglycemic effect, but the actual mechanism remains unclear. This study explored the hypoglycemic mechanism of aqueous extracts of CM in normal Wistar rats. First, the optimal dose of CM for lowering plasma glucose and insulin secretion was tested. Further, atropine and hemicholinium-3 (HC-3) were injected and a western blot was used to investigate insulin signaling. It was found that 10 mg/kg CM extracts had a stronger hypoglycemic effect than a higher dose (100 mg/kg); therefore, a dose of 10 mg/kg was used in subsequent experiments. In normal rats, CM extracts decreased plasma glucose by 21.0% and induced additional insulin secretion by 54.5% after 30 min. When atropine or HC-3 was injected, CM induced a hypoglycemic effect, but the enhancement of insulin secretion was blocked. By western blotting, significant increases in the insulin receptor substrate 1 (IRS-1) and glucose transporter 4 (GLUT-4) were observed after CM feeding. However, the elevation of these signaling proteins was abolished by atropine or HC-3. Taken together, these findings indicate that CM can lower plasma glucose via the stimulation of insulin secretion and cholinergic

activation involved in the hypoglycemic mechanism of normal Wistar rats. Copyright © 2012 John Wiley & Sons, Ltd.

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